

TUNG T. NGO

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RESEARCH INTERESTS

Human-Robot Interaction, VLA, Human-Centered Development, Reinforcement Learning, Computer Vision, SLAM

EDUCATION

Ph.D. in Computer Science, Technological University Dublin (funded by ADAPT Center) 9/2023 - Now

M.Sc. (GPA: 9.32/10.0) in Mechatronics, Hanoi University of Science and Technology (funded by VINIF) 10/2020 - 9/2022
Thesis: R&D of SLAM algorithm for autonomous intelligent robots applying AI

B.S.E. (GPA: 3.25/4.0) in Mechatronics Engineering, Hanoi University of Science and Technology 9/2015 - 8/2020
Highest entrance score of the Mechatronics Talented Program

EXPERIENCE

ADAPT Center 9/2023 - Now
PhD Researcher Dublin, Ireland

Advisors: [Dr. Robert Ross](#), [Dr. Conor Mc Ginn](#), and [Dr. Emma Murphy](#)

Project [1] [2]: Enhancing Multilingual Communication in Healthcare through Embodied Medical Interpreter Agents

- Conduct user studies to investigate user needs and feedback to recommend designs for medical interpreter robots;
- Apply foundation models (LLMs, VLMs, VLA) to enhance robotic perception in healthcare settings;
- Research reinforcement learning from human feedback algorithms to generate informative humanoid robot gestures.

School of Computer Science, TU Dublin 3/2024 - Now
High-Performance Computing Clusters Administration Assistant Dublin, Ireland

- Administer multi-node HPC clusters and support users in job execution, troubleshooting, and performance optimization.

College of Engineering and Computer Science, VinUniversity 9/2022 - 7/2023
Research and Teaching Assistant Hanoi, Vietnam

Advisors: [Dr. Truong Do](#), and [Dr. Cuong Do](#)

TA Courses: Intelligent Physical Systems, Mechatronics, Mechanical Synthesis, Mechanics of Engineering Materials, Thermodynamics, Computer System Programming, and Dynamics.

Project [3]: MedBot: A Hospital-assistant Robot with Active SLAM, Sensor Fusion, and Chatbot Capabilities

- Mentored an undergrad team developing on-device natural language navigation for robots in challenging indoor scenarios.

VinUni-Illinois Smart Health Center, VinUniversity 3/2022 - 8/2022
Research Assistant Hanoi, Vietnam

Advisors: [Dr. Hieu Pham](#), and [Prof. Minh Do](#)

- Proposed a method using deep reinforcement learning to design implants for varied skull defects.

Autonomous Intelligent Robotics Lab, Hanoi University of Science and Technology 5/2018 - 8/2022
Research Assistant Hanoi, Vietnam

Advisor: [Assoc. Prof. Dr.-Ing. Xuan-Ha Nguyen](#)

Project [4] [5]: Deep Learning in Computer Vision and GraphSLAM for long-term autonomous vehicle applications

Collaborator: [Autonomous Intelligent Systems Lab](#), University of Freiburg, Germany

- Researched probabilistic navigation algorithms: GMapping, AMCL, EKFSLAM, and GraphSLAM;
- Proposed and evaluated a lightweight traffic landmark extraction method combining object detection and depth estimation, increasing accuracy and reducing computational cost.

Project [6] [7] [8]: R&D of SLAM algorithms for autonomous intelligent robots in logistics and services

- Designed software and hardware systems of an intelligent service mobile robot (AIR-HUST);
- Enhanced SLAM and obstacle avoidance through multi-layer sensor fusion (IR, LIDAR, and sonar) approach.

CMC Institute of Science and Technology, CMC Corporation 5/2018 - 7/2020
R&D Intern Hanoi, Vietnam

Advisor: [Assoc. Prof. Dr.-Ing. Xuan-Ha Nguyen](#)

- Developed a reception robot ([C-Bot](#)) based on TurtleBot2: visitor engagement, guided tours, and interactive showcase.

PUBLICATIONS

List of publications on [Google Scholar](#).

- [1] **T. T. Ngo**, E. Murphy, and R. Ross, "Vision-Language System using Open-Source LLMs for Consent and Instruction Gestures in Medical Interpreter Robots," in Companion Proc. 21st ACM/IEEE Int. Conf. on Human-Robot Interaction, Edinburgh, UK, 2026. doi: 10.1145/3776734.3794357.
- [2] **T. T. Ngo**, E. Murphy, C. McGinn, and R. Ross, "Body-Oriented Gesture Generation System for Medical Interpreter Robots Based on Reinforcement Learning from Human Feedback," in Int. Conf. AI in Healthcare Collection of Short Abstr., Cambridge, UK, 2025. doi: 10.5281/zenodo.17037425.
- [3] **T. T. Ngo** et al., "Development of a Cost-effective On-device Natural Language Command Navigation System for Mobile Robots in Challenging Indoor Scenarios," in Proc. 2025 IEEE/SICE Int. Symp. Syst. Integration (SII), Munich, Germany, 2025, pp. 1255-1259, doi: 10.1109/SII59315.2025.10870597.
- [4] H. X. Nguyen, **T. T. Ngo**, T. V. Nguyen, A. D. Pham, and T. D. Nguyen, "An efficient approach for traffic sign detection, classification, and localization applied for autonomous intelligent vehicles," J. Modern Phys. Lett. B, Feb. 2023, doi: 10.1142/S0217984923400110.
- [5] H. X. Nguyen, **T. T. Ngo**, and A. D. Nguyen, "Development of real-time traffic-object and traffic-sign detection models applied for autonomous intelligent vehicles," J. Sci. Technol. Smart Syst. Devices, vol. 32, pp. 17-24, Jan. 2022, doi: 10.51316/jst.155.ssad.2022.32.1.3.
- [6] H. X. Nguyen, **T. T. Ngo**, and H. V. Nguyen, "Development of an autonomous intelligent mobile robot based on AI and SLAM technology," in Proc. Int. Conf. Intell. Syst. Netw. 2021, pp. 319-326, doi: 10.1007/978-981-16-2094-2_40.
- [7] H. X. Nguyen, H. V. Nguyen, **T. T. Ngo**, and A. D. Nguyen, "Improvement of Control Algorithm for mobile robot using multi-layer sensor fusion," Vietnam J. Sci. Technol., vol. 59, no. 1, pp. 110-119, Feb. 2021, doi: 10.15625/2525-2518/59/0/15301.
- [8] H. X. Nguyen, H. V. Nguyen, and **T. T. Ngo**, "A new landmark detection approach for SLAM algorithm applied in mobile robot," J. Sci. Technol. Tech. Univ., vol. 146, pp. 31-36, Nov. 2020, doi: 10.51316/30.7.6.

HONORS AND AWARDS

Top 15 Huawei Ireland Tech Arena: An LLM Efficiency Challenge Huawei Ireland Research Centre	2025
Domestic Master Scholarship (full tuition fee and monthly stipend) Vingroup Innovation Foundation, Vingroup Big Data Institute	2020
1st Prize in Student Research Competition Hanoi University of Science and Technology	2020
4th Prize in the 2017 Blitz Research Competition Vietnam Summer School of Science, Rencontres du Vietnam	2017
Student with five good merits (morality, studying, physical training, volunteer, integration) Vietnam National Union of Students of Hanoi University of Science and Technology	2016-2019
FYT Scholarship for Outstanding Students (top 25 nationwide) FPT Center for Young Talents, FPT Corporation	2015-2017

SKILLS

Programming languages	Python, C/C++, Matlab, XML, Assembly
Frameworks	PyTorch, Scikit-Learn Gym, Mujoco PEFT, TRL ROS, NAOqi CUDA, Shurm
User study	experimental design, hypothesis testing, semi-structured interview, qualitative data analysis

SOCIAL ACTIVITIES

- EUT+ Robotics Research Group 2025
- [Robotics Ambassador - The Construct](#) 2023 - 2025
- Instructor for AI in Robotics - [STEAM for Vietnam](#) 2024 - 2025
- Top 12 Vietnam (full scholarship) - [NUS Enterprise Summer Program in Entrepreneurship](#), Singapore 2019
- Top 25 HUST (fully funded) - TFI Specialists' Community Action and Leadership Exchange at Temasek Polytechnic, Singapore and HUST, Vietnam 2018